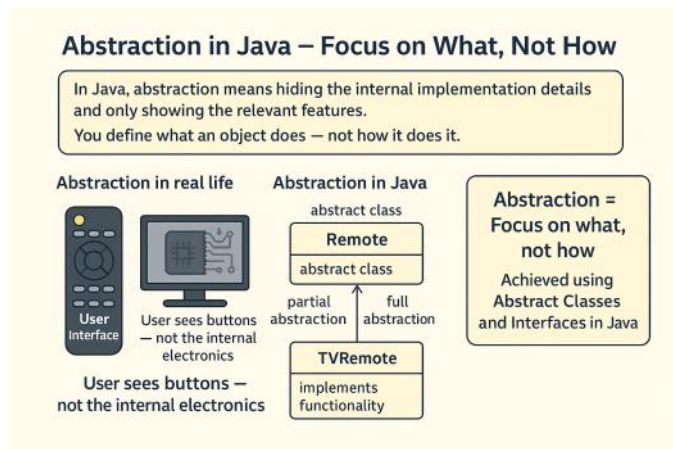


Abstraction in Java



```
// Abstract class
abstract class Payment {
    // Abstract method
    public abstract void makePayment();
}

// Concrete class for Credit Card
class CreditCardPayment extends Payment {
    @Override
    public void makePayment() {
        System.out.println("Processing payment via Credit Card...");
        // Imagine logic here to connect to credit card gateway
    }
}

// Concrete class for UPI
class UPIPayment extends Payment {
    @Override
    public void makePayment() {
        System.out.println("Processing payment via UPI...");
        // Imagine logic here to verify UPI ID and complete payment
    }
}

// Main class to test abstraction
public class PaymentTest {
    public static void main(String[] args) {
        Payment payment1 = new CreditCardPayment();
        payment1.makePayment(); // Output: Processing payment via Credit Card...

        Payment payment2 = new UPIPayment();
        payment2.makePayment(); // Output: Processing payment via UPI...
    }
}
```

```
// Interface that defines the abstraction
interface Vehicle {
    void start();
    void accelerate();
    void stop();
}

// Concrete class that implements the interface
class Car implements Vehicle {
    public void start() {
        System.out.println("Car started with key ignition.");
    }

    public void accelerate() {
        System.out.println("Car is accelerating...");
    }

    public void stop() {
        System.out.println("Car has stopped.");
    }
}
```

```
// Another class that implements the same interface
class ElectricScooter implements Vehicle {
    public void start() {
        System.out.println("Electric scooter started with button.");
    }

    public void accelerate() {
        System.out.println("Electric scooter is zooming silently...");
    }

    public void stop() {
        System.out.println("Electric scooter has stopped.");
    }
}
```

```
// Main class to demonstrate abstraction
public class AbstractionDemo {
    public static void main(String[] args) {
        Vehicle myCar = new Car();
        myCar.start();
        myCar.accelerate();
        myCar.stop();

        System.out.println();

        Vehicle myScooter = new ElectricScooter();
        myScooter.start();
        myScooter.accelerate();
        myScooter.stop();
    }
}
```